

## Partial Fractions 2

Pre-Uni Maths for Sciences A5.2

Subject &amp; topics: Maths | Algebra | Manipulation      Stage &amp; difficulty: A Level P3

The function  $\frac{w+2}{(w-1)(w+1)(2w+1)}$  can be written as  $\frac{A}{(w-1)} + \frac{B}{(w+1)} + \frac{C}{(2w+1)}$ . Using the substitution method find the constants  $A$ ,  $B$  and  $C$ .

**Part A****Find  $A$** Find the constant  $A$ .

The following symbols may be useful: A

**Part B****Find  $B$** Find the constant  $B$ .

The following symbols may be useful: B

Part C

Find  $C$

Find the constant  $C$ .

The following symbols may be useful:  $c$

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**STEM SMART Double Maths 26 - Partial Fractions, Proof & Method of Differences**

## Improper Partial Fractions 2

Pre-Uni Maths for Sciences A5.4

**Subject & topics:** Maths | Algebra | Manipulation      **Stage & difficulty:** A Level P3

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Express  $\frac{16x^3 + 36x^2 + 2x - 25}{4x^2 + 12x + 9}$  as partial fractions.

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## Quadratic Partial Fractions 1

**Subject & topics:** Maths | Algebra | Manipulation      **Stage & difficulty:** Further A P2

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Express  $\frac{5x^2 - 7x + 8}{(x - 2)(x^2 + 3)}$  as partial fractions.

The following symbols may be useful: x

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# Proof and Odd Perfect Numbers

Subject & topics: Maths | Number | Arithmetic
 Stage & difficulty: A Level P3

The **proper divisors** of a number are those factors which are not equal to the number itself. For the number 6,

- The divisors of 6 are 1, 2, 3 and 6.
- The proper divisors of 6 are 1, 2 and 3.

The number 6 is an example of a **perfect number**. A perfect number is a number for which the sum of its proper divisors is equal to the number itself. For the number 6,

$$1 + 2 + 3 = 6$$

In this question you will use proof by contradiction to show that an odd perfect number cannot be a square number.

## Assumption:

We will assume that there is an odd perfect number,  $n$ , that is also a square number. Then  $n = m^2$ , where  $m$  is an integer.

Part A
Reasoning: odd and even factors

- An even number multiplied by an even number is always an  number.
- An even number multiplied by an odd number is always an  number.
- An odd number multiplied by an odd number is always an  number.

Therefore, as  $n$  is an  number, the factors of  $n$  can only be  numbers.

Items:

Part B

Reasoning: sum of proper divisors

As  $n = m^2$ ,  $m$  is a factor of  $n$ .

Consider another factor of  $n$ . Call this factor  $p$ . As  $p$  is a factor of  $n$ ,  $q =$  is also a factor of  $n$ . As  $n = m^2$ ,  $q = \frac{m^2}{p}$ . Hence,

- If  $p < m$ ,  $q$    $m$ .
- If  $p > m$ ,  $q$    $m$ .

Therefore, with the exception of  $m$ , the factors of  $n$  occur in pairs. One factor in the pair is smaller than  $m$ , and the other factor is larger than  $m$ . Including  $m$ , the total number of factors of  $n$  is therefore an  number.

For any value of  $n$ , one of the factor pairs is 1 and  $n$ . The number of proper divisors (factors other than  $n$  itself) is therefore an  number. As we have shown in part A that all of the factors of  $n$  are  numbers, the sum of the proper divisors of  $n$  is therefore an  number.

Items:

even

$\frac{n}{p}$

<

$\frac{p}{n}$

>

odd

## Part C

### Conclusion

Our starting assumption was that  $n$  is an odd perfect number and also a square number.

The definition of a perfect number means that the sum of the proper divisors of  $n$  is equal to . The sum of the proper divisors must therefore be an  number.

However, in part B we have shown that if  $n$  is an odd number which is also a square number, the sum of the proper divisors has to be an  number.

Therefore, we have arrived at a contradiction. We conclude that there are no odd perfect numbers that are also square numbers.

Items:

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## Proof Applied to Surface Areas

**Subject & topics:** Maths | Number | Arithmetic      **Stage & difficulty:** A Level P3

Consider a sphere with a radius  $r$  cm, where  $r$  is a rational number. Using proof by contradiction, show that the side length of a cube with the same surface area cannot also be a rational number of cm.

Once you have worked out a proof yourself, consider the model proof below. Drag and drop answers into the blank spaces to complete the proof. You may use the same answer more than once.

### Assumption:

Consider a sphere of radius  $r$  cm, where  $r$  is a rational number. Let the side length of a cube with the same surface area as the sphere be  $a$  cm. Assume that  $a$  is a rational number, in which case  $a = \frac{b}{c}$ , where  $b$  and  $c$  are integers with no common factor.

### Reasoning:

The surface area of the sphere is . Because  $r$  is a rational number,  $r = \frac{p}{q}$ , where  $p$  and  $q$  are integers with no common factor. Hence, the surface area of the sphere may be written as .

The surface area of the cube is . Using  $a = \frac{b}{c}$ , the surface area may be written as .

The surface area of the sphere and the cube are equal. Hence,  $4\pi \left(\frac{p}{q}\right)^2 = 6 \left(\frac{b}{c}\right)^2$ . Re-arranging this equation to give an expression for  produces .

As  $b$ ,  $c$ ,  $p$  and  $q$  are all integers, this expression implies  must be  number. However,  $\pi$  is not  number.

### Conclusion:

The assumption that  $a$  is rational has resulted in a contradiction. Hence, the assumption cannot be true. Therefore, the side length of a cube with the same surface area as a sphere of radius  $r$  cm, where  $r$  is a rational number, cannot be a rational number of cm.

Items:

$6 \left(\frac{b}{c}\right)^2$

$6a^2$

$a^3$

$\pi = \frac{3b^2q^2}{2c^2p^2}$

$4\pi r^2$

an irrational

a rational

$4\pi \left(\frac{p}{q}\right)^2$

a real

$\pi = \frac{3b^2p^2}{2c^2q^2}$

$$\frac{3b^2q^2}{2c^2p^2}$$

$$\pi$$

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## Partial Fractions Applied to Other Functions

Pre-Uni Maths for Sciences A5.5, A5.6 & A5.7

**Subject & topics:** Maths | Algebra | Manipulation      **Stage & difficulty:** A Level C3

Express the following functions in partial fraction form.

### Part A

#### A trigonometric function

Express the function  $\frac{\cos y}{(\cos y + 1)(2 \cos y + 1)}$  in the form  $\frac{A}{\cos y + 1} + \frac{B}{2 \cos y + 1}$ , where  $A$  and  $B$  are constants.

The following symbols may be useful:  $\cos()$ ,  $\sin()$ ,  $\tan()$ ,  $y$

### Part B

#### An exponential function

Express the function  $\frac{e^{2x} + 5}{(e^x - 1)(e^x - 2)(e^x - 3)}$  in the form  $\frac{A}{e^x - 1} + \frac{B}{e^x - 2} + \frac{C}{e^x - 3}$ , where  $A$ ,  $B$  and  $C$  are constants.

The following symbols may be useful:  $e$ ,  $x$

A logarithmic function

Express the function  $\frac{5 \ln z + 20}{(\ln z)^2 + \ln(z^2) + 1}$  in the form  $\frac{A}{(\ln z + 1)^2} + \frac{B}{\ln z + 1}$ , where  $A$  and  $B$  are constants.

The following symbols may be useful:  $\ln()$ ,  $\log()$ ,  $z$

## Force From Electric Dipole

**Subject & topics:** Physics | Fields | Electric Fields      **Stage & difficulty:** A Level C3

An electric dipole consists of two charges that are equal in size but opposite in sign, with a separation between them. The diagram below shows an electric dipole PQ. P has charge  $-q$  and Q has charge  $+q$ , and the separation between P and Q is  $2a$ . Another charge, S, is near to the dipole. S is in line with the axis of the dipole and a distance  $r$  from the dipole's centre.



**Figure 1:** An electric dipole PQ and a charge S.

The resultant force on charge S is the sum of the force on S from P and the force on S from Q. For a particular value of  $q_S$ , the resultant force is given by the expression

$$F_{\text{res}} = \frac{-3q^2}{4\pi\epsilon_0} \frac{ar}{(r^2 - a^2)^2}$$

where  $\epsilon_0$  is a constant. In this question you will use partial fractions to split  $F_{\text{res}}$  into two terms, and hence find an expression for  $q_S$  in terms of  $q$ .

## Part A

### Splitting into terms - $A$

In general, a rational function with a denominator of  $4\pi\epsilon_0(r^2 - a^2)^2$  would produce four terms when written in terms of partial fractions:

$$\frac{A}{4\pi\epsilon_0(r+a)^2} + \frac{B}{4\pi\epsilon_0(r-a)^2} + \frac{C}{4\pi\epsilon_0(r+a)} + \frac{D}{4\pi\epsilon_0(r-a)}$$

However, if the expression for  $F_{\text{res}}$  is written in terms of partial fractions, it turns out that two of the coefficients ( $C$  and  $D$ ) are both 0.

Write the expression for  $F_{\text{res}}$  in the form  $\frac{A}{4\pi\epsilon_0(r+a)^2} + \frac{B}{4\pi\epsilon_0(r-a)^2}$ , where  $A$  and  $B$  are constants which depend on  $q$ .

Enter your expression for  $A$ .

The following symbols may be useful: A, a, pi, q, varepsilon\_0

## Part B

### Splitting into terms - $B$

Write the expression for  $F_{\text{res}}$  in the form  $\frac{A}{4\pi\epsilon_0(r+a)^2} + \frac{B}{4\pi\epsilon_0(r-a)^2}$ , where  $A$  and  $B$  are constants which depend on  $q$ .

Enter your expression for  $B$ .

The following symbols may be useful: B, a, pi, q, varepsilon\_0

Part C

Finding  $q_S$

The force between two particles with electric charges  $q_1$  and  $q_2$  separated by a distance  $d$  is given by

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 d^2}$$

where  $\epsilon_0$  is a constant.

Using your answers to parts A and B, or otherwise, find an expression for the charge on S,  $q_S$ , in terms of  $q$ .

The following symbols may be useful: a, pi, q, q\_S, varepsilon\_0

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## Series: Method of Differences 2i

**Subject & topics:** Maths    **Stage & difficulty:** Further A P2

### Part A

$$(r + 2)! - (r + 1)!$$

Show that  $(r + 2)! - (r + 1)! = f(r) \times r!$  where  $f(r)$  is a function to be found.

What is  $f(r)$ ?

The following symbols may be useful:  $r$

### Part B

#### Expression for a series

Hence find an expression, in terms of  $n$ , for

$$2^2 \times 1! + 3^2 \times 2! + 4^2 \times 3! + \dots + (n + 1)^2 \times n!$$

Your answer can be written as  $g(n)! - 2$ .

What is  $g(n)$ ?

The following symbols may be useful:  $n$

Part C

Convergence

State, giving a brief reason, whether the series

$$2^2 \times 1! + 3^2 \times 2! + 4^2 \times 3! + \dots$$

converges. Fill in the gaps in the argument (you can use an item more than once).

We can express this series as a summation as . This is the limit of the partial sum  as  $n \rightarrow$  .

From Part A we can write the partial sum as , and from Part B we know that the partial sum evaluates to .

As  $n \rightarrow \infty$ , the partial sum tends to , so the series  converge.

Items:

$\sum_{r=1}^{\infty} (r+1)^2 r!$

$\sum_{r=1}^{\infty} r^2 (r+1)!$

$\sum_{r=1}^n (r+1)^2 r!$

$\sum_{r=1}^n r^2 (r+1)!$

0

1

$\infty$

$\sum_{r=1}^n [(r+2)! - (r+1)!]$

$\sum_{r=1}^n [(r+1)! - r!]$

$(n+2)! - 2$

$(n+1)! - 1$

does

does not

Adapted with permission from UCLES, A Level, January 2007 , Paper 4725, Question 8.

## Series: Method of Differences 1i

Subject & topics: Maths      Stage & difficulty: Further A P2

Part A

Rewriting a fraction

Express  $\frac{2}{r} - \frac{1}{r+1} - \frac{1}{r+2}$  as a single fraction.

The following symbols may be useful: r

Part B

Sum of a series

Hence find an expression, in terms of  $n$ , for

$$\sum_{r=1}^n \frac{3r + 4}{r(r + 1)(r + 2)}.$$

The following symbols may be useful: n

Part C

Limit as  $n \rightarrow \infty$

Hence write down the value of

$$\sum_{r=1}^{\infty} \frac{3r+4}{r(r+1)(r+2)}.$$

Part D

Solve for  $N$

Given that

$$\sum_{r=N+1}^{\infty} \frac{3r+4}{r(r+1)(r+2)} = \frac{7}{10}$$

find the value of  $N$ .

Adapted with permission from UCLES, A Level, January 2008 , Paper 4725, Question 10

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